

Seam Carving for Content-Aware Image Resizing

N8EN11D Projet Traitement d'image / Image Processing project



Figure 1: On the left, the original photo. In the middle image, the red lines represent the pixel “seams” that, if removed, would leave the photo relatively intact. The image on the right shows what the photo would look like after removing the seams.

1 Presentation

The goal of this project is to read, understand, and implement the algorithm described in the paper “Seam Carving for Content-Aware Image Resizing” [1]. The Seam Carving algorithm is an innovative image resizing technique that preserves important information while adjusting the image size non-uniformly. The algorithm is based on the concept of a “seam”, which is a line of pixels traversing the image vertically or horizontally. Each seam represents a path through the image, and its selection is based on the notion of pixel energy. The energy of a pixel is a measure of its importance in the image and is calculated using local brightness or color gradients. The main step of the algorithm is to find and remove the seams with the lowest energy to resize the image.

2 Description of the project

1. **Reading and Understanding:** Carefully read the paper to understand the principles and mechanisms of the “Seam Carving” algorithm.
2. **Implementation:** Implement the method in a programming language of choice (C/C++, Python, Matlab, etc.).
 - Ensure the code is **well-structured and modular**, with proper decomposition into functions.
 - Your implementation should sufficiently generic to use **different Image Energy Functions as proposed** in Sec 3.2 of the paper. As for the segmentation energy, you can consider using meanshift or superpixels, as seen during the other lab sessions.
 - It should be easy to run the code with **any given image and user-defined parameters**, such as the type of energy to use, the orientation of the seams (horizontal, vertical) *etc.*
 - *While not required, optimizing the implementation with **parallelization techniques** (e.g., **OpenMP** for C++, **parfor** for MATLAB, etc.) could be a **bonus**— but only if you are familiar with these techniques, as the main focus should remain on the implementation and evaluation.*

3. **Experimentation:** Test the algorithm with different combinations of energy functions over images with different types of content. Examples of datasets you can use are the “Kodak dataset” [2] or the “RetargetMe Dataset” [5] that is specifically designed for Content-Aware resizing algorithms. You can also explore the applications proposed in the paper (*c.f.* Sect 4, object removal, retargeting with optimal seam-order *etc.*).
4. **Evaluation of Results:** Evaluate the results of Seam Carving using **quantitative metrics**. Examples of metrics that could be interesting to use are:
 - **Gradient Magnitude Preservation:** Since Seam Carving removes low-energy seams, it should preserve high-energy regions (*e.g.*, edges, important textures). Compute the image gradient using Sobel or Canny edge detection and analyze:
 - Mean Gradient Magnitude: A significant drop in gradient strength suggests a loss of important edges.
 - Edge Density: The number of edge pixels before and after resizing. A strong reduction may indicate over-smoothing.
 - **Entropy Preservation:** Entropy measures the richness of texture and details in the image. A large entropy drop suggests details loss due to excessive seam removal.
 - **Saliency Preservation:** Saliency refers to the parts of an image that attract human attention the most. These are usually high-contrast areas, edges, faces, objects, or regions of interest that stand out from the background. In image processing, saliency detection is a method to identify and highlight the most visually important parts of an image. It is often used in tasks such as object detection, image compression, and content-aware editing. Since seam carving removes pixels adaptively, it’s crucial to ensure that important regions (*e.g.*, faces, text, key objects) are not distorted or lost. By comparing saliency maps before and after resizing, we can evaluate whether the algorithm preserves the most visually significant content. You can use some existing implementations to compare the saliency (*e.g.*, OpenCV [3], Matlab [4]).

References

- [1] Shai Avidan and Ariel Shamir. “Seam carving for content-aware image resizing”. In: *ACM SIGGRAPH 2007 Papers*. SIGGRAPH ’07. San Diego, California: Association for Computing Machinery, 2007, 10–es. ISBN: 9781450378369. DOI: 10.1145/1275808.1276390. URL: https://graphics.cs.cmu.edu/courses/15-463/2013_fall/hw/proj3-seamcarving/imret.pdf.
- [2] Kodak. *Kodak Lossless True Color Image Suite*. URL: <https://r0k.us/graphics/kodak/>.
- [3] OpenCV. *Saliency’s module*. URL: https://docs.opencv.org/4.2.0/d8/d65/group__saliency.html.
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- [5] Michael Rubinstein et al. “A Comparative Study of Image Retargeting”. In: *ACM Transactions on Graphics (Proc. SIGGRAPH ASIA)* 29.6 (2010), 160:1–160:10. URL: <https://people.csail.mit.edu/mrub/retargetme/download.html#dataset>.